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Experimental investigation and thermodynamic modeling of the Gd–Si system

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Abstract

The Gd–Si system has been investigated via experiments and thermodynamic modeling. In the experimental part, alloys were prepared by arc melting pure Gd and Si slugs and annealing at high temperature. The annealed alloys were then investigated by thermal analysis, X-ray diffraction and scanning electron microscopy. Except for the compound with AlB_2 structure, the existence of all other compounds was confirmed. In our study, instead of Gd_2Si_3, the compound with AlB_2 structure was found to be Gd_3Si_5. Although most of the reported invariant reaction temperatures are in good agreement with the present data, the convex liquidus in the composition range of 55–65 at.% Si was not found. The available phase equilibrium and thermodynamic data for the Gd–Si system were used in optimization of thermodynamic parameters by WinPhad software. The calculated phase equilibria and thermodynamic properties were compared to the corresponding experimental data and good agreement was found.
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Keywords: Gd–Si phase diagram; Thermal analysis; X-ray diffraction; Scanning electron microscopy; Thermodynamic modeling

1. Introduction

The binary and ternary gadolinium-based compounds of Si and Ge are of current interest due to their electric [1], magnetic [2] and giant magnetocaloric [3] properties that results from magnetostructural transformations. Recently, intensive studies were carried out concerning the magnetocaloric properties of $Gd_5(Ge_{1-x}Si_x)_4$ and to understand some of the fundamental aspects of the behavior of these compounds, investigation of single crystalline properties were desired. Knowledge of phase diagrams and thermodynamic properties of the alloys system often has significant impact on single crystal synthesis, and as a matter of course we evaluated the available phase equilibrium of the ternary as well as the binary end members. While assessing the ternary phase equilibria of the Gd–Si–Ge in the vicinity of the Gd_5X_4 compounds, we noticed considerable differences in the literature for Gd–Si phase diagram especially at higher Si contents. In particular, there are discrepancies reported in the region of 60–67 at.% Si concerning both the existence of various compounds between 60 and 67% Si and secondly, the oddly shaped liquidus surface is not consistent with thermodynamic rules or phase diagram construction. It is this latter point that initially prompted us to investigate the phase equilibria in the Gd–Si system. While there are several experimental determinations of the phase diagram available in the literature, to the best our knowledge, no thermodynamic calculations have been done so far. Therefore, a thermodynamic description for the Gd–Si system is desired by using both experiment and thermodynamic modeling.

The Gd–Si binary system was first evaluated by Gokhale and Abbaschian in 1989 [4] and was based on results from several studies carried out over limited composition and/or temperature range. Two years later, this system was comprehensively investigated by Eremenko et al. [5] using metallography, electron microprobe analysis and chemical analysis, differential thermal analysis (DTA), and X-ray diffraction (XRD). Among the five intermediate compounds, GdSi is the only silicide that melts congruently at 1830 °C; the other four silicides, Gd_5Si_3, Gd_5Si_4, Gd_2Si_3, and $GdSi_{2-x}$ incongruently melt at 1700, 1725, 1625 and 1590 °C, respectively. Two eutectic reactions were reported: one between (Gd) and Gd_5Si_3 at 1080 °C and the other between $GdSi_{2-x}$ and (Si) at 1230 °C, respectively. The crystal structure of Gd_5Si_3 is of the Mn_5Si_3 hexagonal type [6]; the structure of Gd_5Si_4 is of the Sm_5Ge_4 orthorhombic type; and the structure of

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the GdSi is of the FeB orthorhombic type. Recently, a new compound Gd_3Si_4 belonging to Ho_3Si_4 type was reported by Roger et al. [7] though they did not succeed to synthesize a pure sample of this compound. There is an agreement that $GdSi_2$ has vacancy on the Si site so that the more accurate formula is $GdSi_{2-x}$; a composition close to $GdSi_{1.4}$ was estimated by Perri et al. [8] based on chemical analyses, X-ray and density measurements but the homogeneity was uncertain. A small range of homogeneity were reported ranging from 64 to 68 at.% Si [9] and 63.2 to 65.4 at.% Si [5], which corresponds to the formula $GdSi_{1.78–1.94}$ [9] and $GdSi_{1.72–1.89}$ [5], respectively; $GdSi_{2-x}$ was also discovered to be dimorphic with a low temperature of orthorhombic structure and a high temperature of the α-$ThSi_2$ [10] tetragonal structure.

For the compound with AlB_2 structure, two stoichiometric compositions were reported at Gd_2Si_3 (60 at.% Si) [5,9] and Gd_3Si_5 (62.5 at.% Si) [11,12], respectively. Our previous study suggested that Gd_3Si_5 is most likely the composition for the phase with the AlB_2 structure [12] since this is true for the neighboring elements on both sides of Gd such as Dy, Er, Ho and Sm. In addition, adding to the confusion surrounding these compounds, the liquidus curve has been oddly constructed [5] in the composition range of 55–65 at.% Si, which is a very uncommon shape for a phase diagram.

The thermodynamic properties of the Gd–Si system have been investigated by several research groups. The enthalpy of formation for Gd_5Si_3 [13,14], Gd_5Si_4 [13], GdSi [13,15], Gd_2Si_3 [13] and $GdSi_{2-x}$ [13,14] were investigated by Usenko et al. [13] by solution drop calorimetry, and Meschel and Kleppa [14,15] by direct-synthesis calorimetry. The Gibbs energy, enthalpy and entropy of formation of the compounds Gd_5Si_3 [16,17], Gd_5Si_4 [16,17], GdSi [16], Gd_2Si_3 [16] and $GdSi_{2-x}$ [16,18] were investigated by Lukashenko et al. [16,18], Sidorko and Polotsokaya, [17] and Lukashenko and Polotskaya [19] by measuring the emf of high-temperature galvanic cells in the temperature range of 600–960 K and the Gibbs energy of formation of Gd_3Si_5 was calculated by Uspenskaya et al. [20]. The enthalpies of mixing of Gd–Si alloys have been measured by high-temperature calorimetry on Si-rich side (94–100 at.% Si) at 1800 K [21], in the composition ranges $0 < x_{Si} < 0.2$ and $0.8 < x_{Si} < 1$, and $0 < x_{Si} < 0.2$ and $0.7 < x_{Si} < 1$ at 1760 K [22,23], and those over the entire composition range at 1770 ± 5 K [24]. The enthalpy of formation for the solid phases and enthalpies of mixing of liquid are all negative.

Therefore, the objective of the present study was to clarify these two issues: the composition of the AlB_2 and the convex nature of liquidus curve. In addition, a thermodynamic description of the Gd–Si system was obtained based on the experiment and thermodynamic data.

2. Experimental procedures

Alloys were prepared from high purity elemental Gd and Si. The Gd was supplied by the Materials Preparation Center [25] at the Ames Laboratory and it was (99.995 wt.%) pure. The major impurities in ppm by weight were as follows: O, 155; C, 49; Si, 36; Cl, 36; S, 34; and single crystal Si with a purity of 99.9999 at.%. All alloys were formed by arc melting appropriate amounts of the constituent elements. Mass losses during melting were less than 0.5%. Each button was then chill cast into a copper mold to facilitate the homogenization of the phase equilibria during subsequent annealing. Each ingot was wrapped with tantalum foil, sealed in quartz tubes filled with 99.998% argon, and homogenized in a furnace at 1100 °C for 2 weeks then furnace cooled.

The microstructure of the annealed samples was investigated by optical microscopy or scanning electron microscopy (SEM). The quantitative analysis of individual phases was done by energy-dispersive spectroscopy (EDS) on a JEOL 7830F Auger Microprobe. An uncertainty less than ±1% was obtained by using GdSi as standards and also removing of the oxide layer on the sample surface by ion sputtering.

The differential thermal analysis investigations were carried out in a Perkin Elmer DTA 7 which can reach a maximum temperature of 1500 °C. The thermocouples were calibrated to the transition temperature of high purity K_2SO_4 and the melting point of Au and the temperature measured were accurate to within ±5 °C. For the measurements, samples weighing 20–60 mg were cycled at heating and cooling rates of 10 °C/min in tantalum crucibles under an Ar atmosphere gettered by Zr at 600 °C with a flow rate of 50 cm^3/min. Due to the weak signal of the polymorphic transition in the silicides, some of the alloys were analyzed in a differential scanning calorimeter (DSC). The thermocouples used in the DSC were calibrated against the melting points of pure In and Al and the accuracy of the temperature was estimated to be within ±2 °C. DSC measurements were carried out on samples wrapped with tantalum foils then crimped in a copper pan. The measurements were conducted between room temperature and 700 °C.

In order to investigate the liquidus and melting ranges of alloys with a composition range from 30 to 65 at.% Si, additional thermal analysis was conducted. Alloys were arc-melted, homogenized and sealed in tantalum crucibles with an embedded thermocouple well. Tungsten–Rhenium Type C thermocouples calibrated to the melting point of platinum were used for temperature measurement. Niobium held in a second crucible was used as a reference. The voltages from the two thermocouples were recorded at the same time during heating and cooling. The liquidus temperature was obtained when the difference in voltages is a maximum from the two thermocouples. Only the data collected from the first heating or cooling were considered since liquid alloy reacts with Ta forming tantalum silicides.

X-ray diffraction studies were conducted in a Philips X'Pert PRO diffractometer with Cu Kα radiation at 45 kV and 40 mA. Scans were typically taken over a 2θ range of 20–90° using a scan step of 0.02°.

3. Thermodynamic modeling

The liquid, bcc (Gd), hcp (Gd) and diamond (Si) phases are treated by a simple substitutional model with the Gibbs energy expressed as following equation in terms of one mole of atoms:

$$G_m^\phi = x_{Gd}\,{}^\circ G_{Gd}^\phi + x_{Si}\,{}^\circ G_{Si}^\phi + RT(x_{Gd}\ln x_{Gd} + x_{Si}\ln x_{Si}) + \Delta^{xs}G_m^\phi \quad (1)$$

in which x_i is the mole fraction of element i (i = Gd, Si), R the gas constant, and G_i^ϕ is the molar Gibbs energy of element i with a structure of ϕ (liquid, bcc, fcc and hcp) in the nonmagnetic state. In the present work, the Gibbs energy functions for Gd and Si are taken from the SGTE-compilation by Dinsdale [26]. $\Delta^{xs}G_m^\phi$ is the excess Gibbs energy with interaction parameters expressed in the Redlich-Kister polynomial [27]:

$$\Delta^{xs}G_m^\phi = x_{Gd}x_{Si}[{}^0L + {}^1L(x_{Gd} - x_{Si}) + {}^2L(x_{Gd} - x_{Si})^2] \quad (2)$$

where ${}^0L$, ${}^1L$ and ${}^2L$ are the interaction parameters, which could be temperature dependent. There are five intermetallic compounds in the Gd–Si binary system, i.e. Gd_5Si_3, Gd_5Si_4, GdSi, Gd_3Si_5 and $GdSi_{2-x}$ (the Gd_3Si_4 is not included since we do not find this phase based on our SEM/XRD results). They all are treated as stoichiometric phases since the homogeneity range of

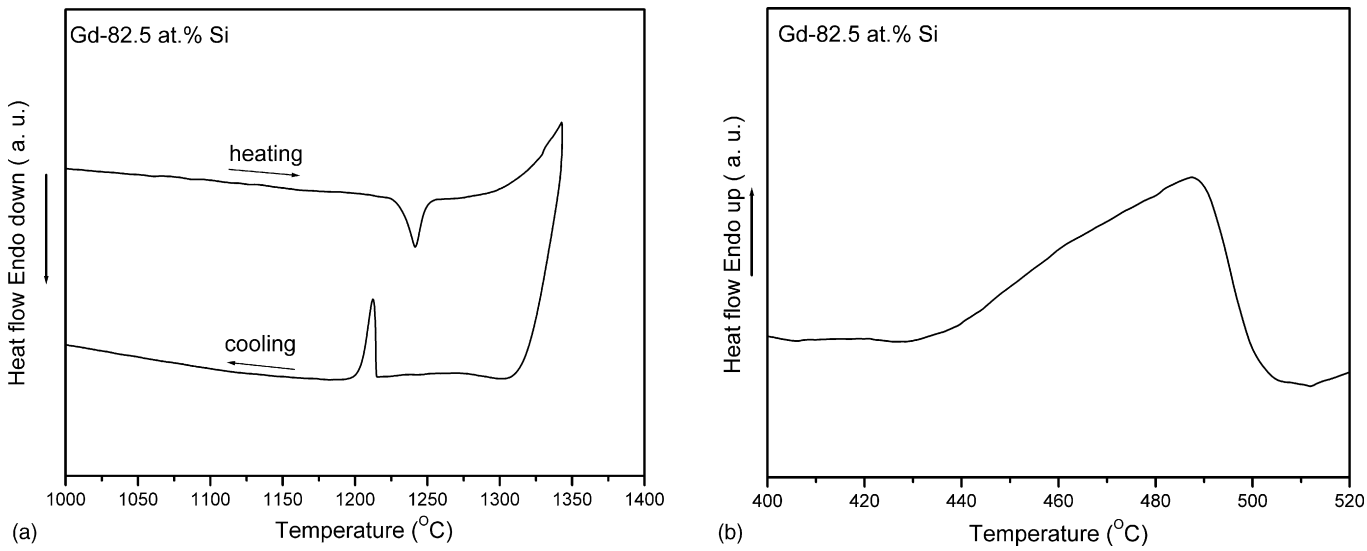

Fig. 1. DTA (a) and DSC (b) traces of alloys Gd–82.5 at.% Si.

$GdSi_{2-x}$ is small. Because the Curie temperatures of all compounds in the Gd–Si system are far below 298.15 K, the magnetic contribution to the Gibbs energy is ignored. For compounds with no heat capacity data, and assuming that the Neumann-Kopp rule holds, their Gibbs energies are expressed by the following equation in terms of 1 mol of atoms: referring to the pure elements in their nonmagnetic reference states:

$$G_{\mathrm{m}}^{\mathrm{Gd}_a\mathrm{Si}_b} = a\,{}^{\circ}G_{\mathrm{Gd}}^{\mathrm{ref}} + b\,{}^{\circ}G_{\mathrm{Si}}^{\mathrm{ref}} + A + BT \quad (3)$$

where ${}^{\circ}G_i^{\mathrm{ref}}$ is the molar Gibbs energy of the pure element i (i = Gd, Si) with its defined reference structure in a nonmagnetic state fro the compilation by Dinsdale [26]. A and B are parameters to be evaluated during the course of optimization.

4. Results and discussion

Fig. 1 shows the DTA and DSC traces of alloys Gd–82.5 at.% Si. The DTA result in Fig. 1(a) indicates a thermal event at 1227 °C on the heating curve for the eutectic reaction liquid ↔ $GdSi_{2-x}$ + Si, which is in good agreement with temperature of 1230 °C reported by Eremenko et al. [5]. A polymorphic phase transformation for $GdSi_{2-x}$ was reported by Eremenko et al. [5] at 500 °C. This phase transformation was hard to detect by DTA due to the weak signal. Therefore, DSC was used to investigate the reaction as shown in Fig. 1(b), indicating a thermal event with an onset temperature and peak temperature at 435 and 487 °C, respectively, which are lower than the reported data of 500 °C [5].

Fig. 2 shows the XRD result of the annealed Gd–82.5 at.% Si alloy, indicating that $GdSi_{2-x}$ and Si are in equilibrium as expected. Using the Si as an internal standard, the $GdSi_{2-x}$ was identified as an orthorhombic structure with a = 4.0072 Å, b = 4.0867 Å, and c = 13.4540 Å, which are in good agreement with the reported results [5].

In order to investigate the existence of the Gd_3Si_5 phase, alloys with compositions of Gd–62.5 at.% Si and Gd–63 at.% Si were used. Fig. 3 shows the DSC results of Gd–63 at.% Si alloy, which has a single thermal event with an onset and peak temperature at 637 and 649 °C, respectively, which are lower than the reported data of 700 °C and no peak was found at 425 °C [5]. The thermal event at around 700 °C was not found in our study for alloys with a silicon content of 62.5 at.% Si or less, indicating that the transition observed near 700 °C in this study is related to the α-$GdSi_{2-x}$ to β-$GdSi_{2-x}$ polymorphic transformation. The difference in the transition temperature for α-$GdSi_{2-x}$ to β-$GdSi_{2-x}$ on either side of the phase boundaries is consistent with reported homogeneity range for $GdSi_{2-x}$ [5].

Fig. 4 shows the XRD result of annealed Gd–62.5 at.% Si (Gd_3Si_5) alloy. Except for the peak near 33°, all other peaks can be indexed as an AlB_2 structure with a = 3.8737 Å and c = 4.1753 Å. Though the lattice parameters are in good agreement with the reported results [5], the composition is not Gd–60 at.% Si as previously reported [5]. Our XRD results of

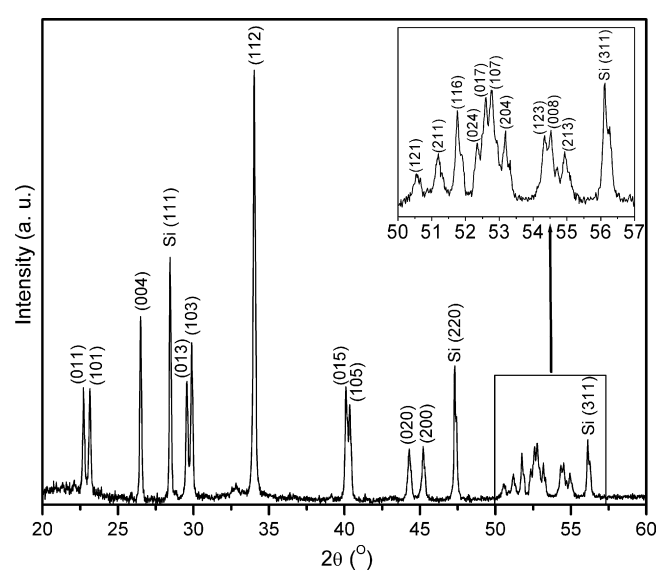

Fig. 2. XRD result of annealed Gd–82.5 at.% Si alloy, indicating that $GdSi_{2-x}$ and Si are in equilibrium.

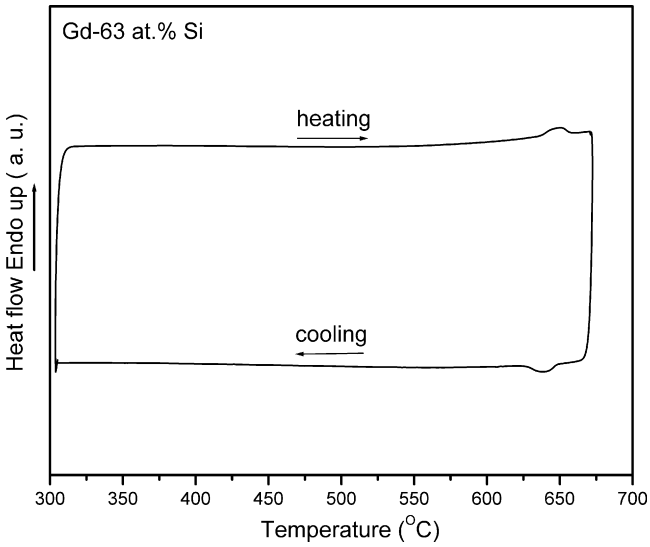

Fig. 3. DSC result of Gd–63 at.% Si alloy shows a thermal event with an onset and peak temperature at 637 and 649 °C, respectively.

alloys Gd–55 at.% Si and Gd–60 at.% Si (Fig. 5) indicate that there is a mixture of two phases consisting of GdSi and Gd_3Si_5, which is in agreement with our back-scatter images of these two alloys shown in Fig. 6, where the light gray phase is GdSi and the dark gray one is Gd_3Si_5. The contrast in the dark gray grains is due to different crystal orientations. XRD and back-scatter images confirm that the phase with the AlB_2 structure has a composition of Gd_3Si_5 not Gd_2Si_3. Recently, Roger et al. [7] reported the existence of compound Gd_3Si_4 with the structure of Ho_3Si_4 even though did not obtain a pure sample of this compound. This is contradict with our present XRD results shown in Fig. 5, which indicates that GdSi and Gd_3Si_5 are in equilibrium and there are no other phases in between.

Optimization of model parameters using selected experimental data was carried out using WinPhad software [28], a computer program developed by CompuTherm (Madison, USA) for the calculation and optimization of thermodynamic parameters for binary systems. This program can accept different types of data, such as any thermodynamic quantities and phase equilibrium data, in the same operation. In this work, all of the optimized parameters are in the form of $A + BT$. The meaning of A and B parameter is that they are the enthalpies and entropies of formation of the phases, respectively. During optimization, the A and B parameters are adjusted to satisfy both the thermodynamic data and phase diagram information within defined limits.

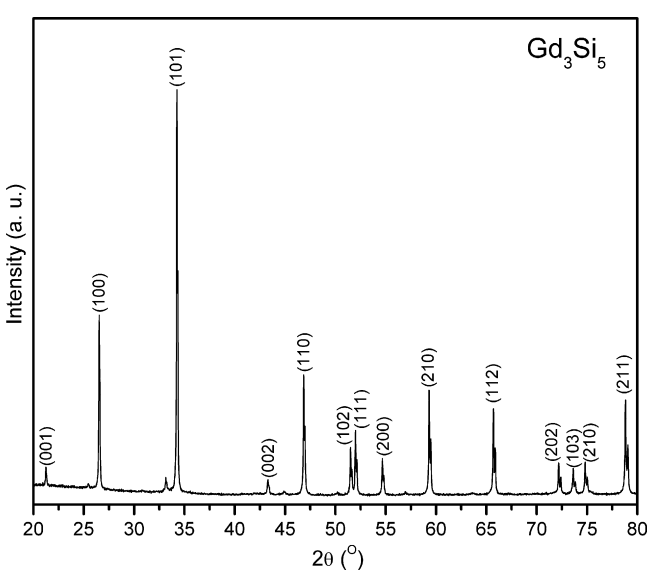

Fig. 4. XRD result of annealed Gd–62.5 at.% Si (Gd_3Si_5) alloy. This phase can be indexed as an AlB_2 structure with $a = 3.8737$ Å and $c = 4.1753$ Å.

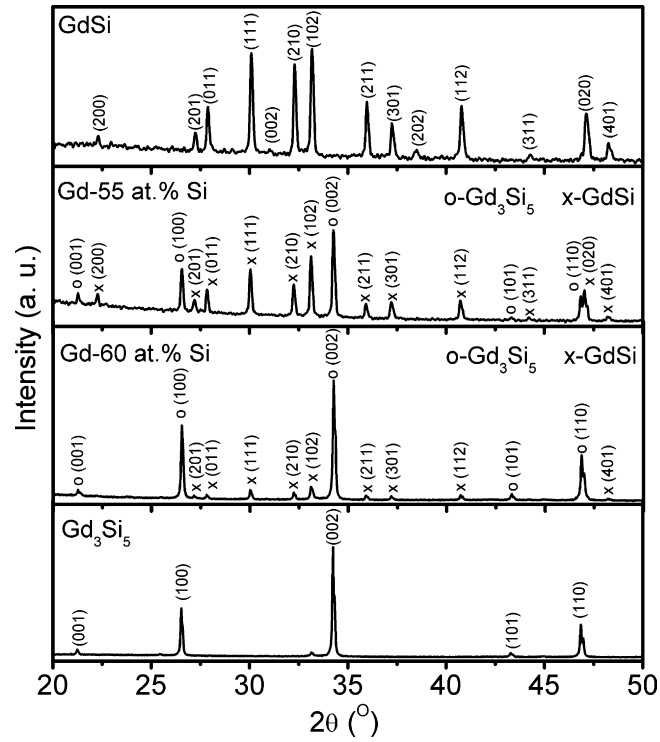

Fig. 5. XRD results of annealed Gd–50 at.% Si (GdSi), Gd–55, 60 at.% Si and Gd–62.5 at.% Si (Gd_3Si_5) alloys indicate GdSi and Gd_3Si_5 are in equilibrium with each other and there are no other phases in between.

Table 1
Optimized thermodynamic model parameters of Gd–Si binary system, in SI units

Liquid	${}^{\circ}L^{\text{liquid}}_{\text{Gd,Si}} = -265000 + 60T$
	${}^{1}L^{\text{liquid}}_{\text{Gd,Si}} = 60000 - 25T$
bcc (Gd)	${}^{\circ}L^{\text{bcc}}_{\text{Gd,Si}} = -5000$
hcp (Gd)	${}^{\circ}L^{\text{hcp}}_{\text{Gd,Si}} = -10000$
Diamond (Si)	${}^{\circ}L^{\text{diamond}}_{\text{Gd,Ni}} = 24411$
Gd_5Si_3	$G^{\text{Gd}_5\text{Si}_3}_{\text{m}} - 0.625\,{}^{\circ}G^{\text{hcp}}_{\text{Gd}} - 0.375\,{}^{\circ}G^{\text{diamond}}_{\text{Si}} = -76893 + 13.19T$
Gd_5Si_4	$G^{\text{Gd}_5\text{Si}_4}_{\text{m}} - 0.5556\,{}^{\circ}G^{\text{hcp}}_{\text{Gd}} - 0.4444G^{\text{diamond}}_{\text{Si}} = -83000 + 14.33T$
GdSi	$G^{\text{GdSi}}_{\text{m}} - 0.5\,{}^{\circ}G^{\text{hcp}}_{\text{Gd}} - 0.5\,{}^{\circ}G^{\text{diamond}}_{\text{Si}} = -87500 + 15.34T$
Gd_3Si_5	$G^{\text{Gd}_3\text{Si}_5}_{\text{m}} - 0.375\,{}^{\circ}G^{\text{hcp}}_{\text{Gd}} - 0.625\,{}^{\circ}G^{\text{diamond}}_{\text{Si}} = -79553 + 14.19T$
$GdSi_{2-x}$	$G^{\text{GdSi}_{2-x}}_{\text{m}} - 0.3333\,{}^{\circ}G^{\text{hcp}}_{\text{Gd}} - 0.6667\,{}^{\circ}G^{\text{diamond}}_{\text{Si}} = -76555 + 14.25T$

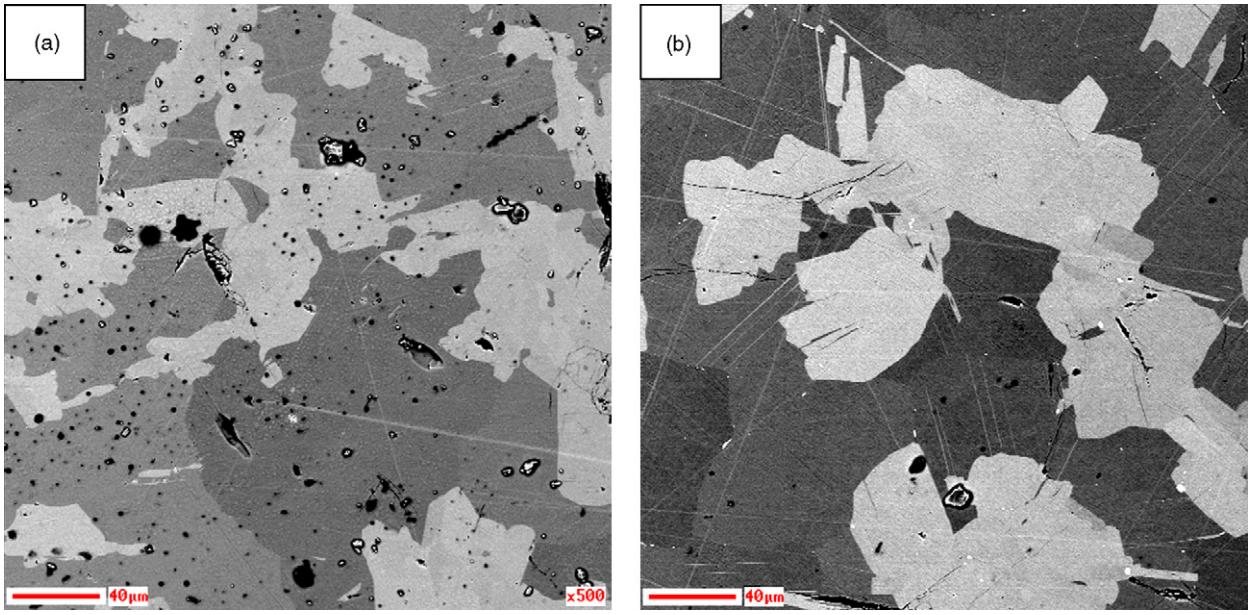

Fig. 6. Back scatter images of Gd–55 at.% Si (a) and Gd–60 at.% Si (b) indicated that there is a mixture of two phases of GdSi and Gd_3Si_5. The light gray phase is GdSi and the dark gray one is Gd_3Si_5.

The obtained optimized model parameters are given in Table 1. Fig. 7 shows the calculated phase diagram using these optimized parameters. Fig. 8 focuses on a portion of the high-temperature region of the calculated phase diagram. It can be seen that the experimental data in the present study are well reproduced by the modeling. However, the reported [5] convex liquidus is not found by either experiments or modeling. The reported [5] liquidus temperature is much lower than that determined in the present study and possibly related to the higher purity of rare-earth starting material used in our study. The calculated and experimental invariant reactions in the Gd–Si system are listed in Table 2.

The enthalpies of formation of the solid phases calculated from the thermodynamic parameters at 298 K and the experimental data are shown in Fig. 9. The calculated results are in reasonable agreement with the reported data [13–15]. The calculated Gibbs energy of solids at 800 K and the reported data by Lukashenko and Polotskaya [18], Sidorko and Polotsokaya [17], and Uspenskaya et al. [20] are shown in Fig. 10. In view of the experimental error in emf, the agreement is reasonable. However, as shown in Figs. 11 and 12, there are large discrepancies between the calculation and experimental data [16,18,19] for the enthalpy and entropy of formation of solids at 800 K, especially on the Gd-rich side. The calculated enthalpies of mixing of the liquid phase at 1770 K together with the experimental data of Kanibolotsky et al. [22,24] and Beloborodova et al. [23] are shown in Fig. 13. The calculated results are in good agree-

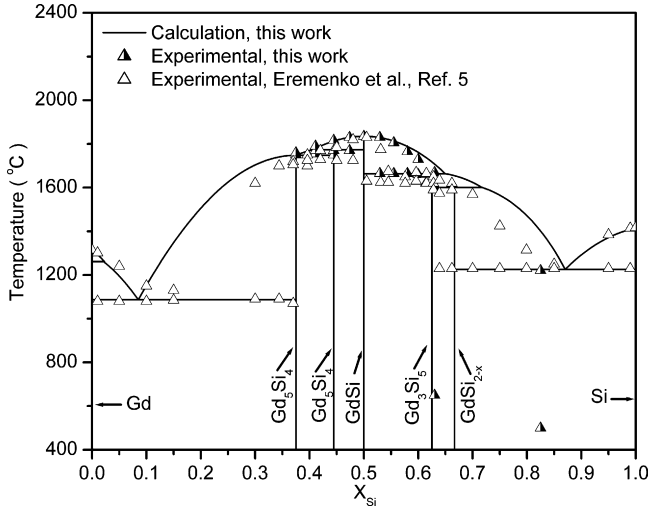

Fig. 7. The calculated Gd–Si binary phase diagram together with the experimental data.

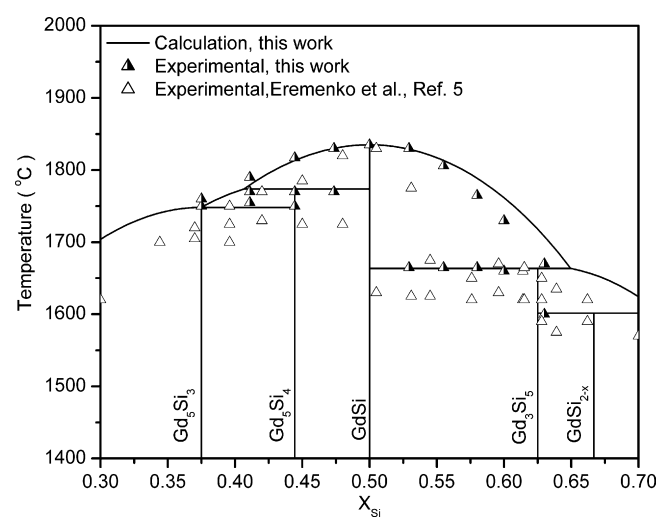

Fig. 8. Enlarged part of the calculated Gd–Si phase diagram with experimental data. The convex liquidus reported in reference was not found by both experiments and calculation.

Table 2
Invariant reactions in the Gd–Ni system

Reaction	Temperature, °C (at.% Si in liquid)	
	Modeling	Experimental
Liquid + bcc (Gd) ↔ hcp (Gd)	1262 (2.12)	
Liquid ↔ Gd_5Si_3 + hcp (Gd)	1087 (8.52)	1088[a], 1080 (∼13.5)[b]
Liquid + Gd_5Si_4 ↔ Gd_5Si_3	1748 (37.46)	1750[a], 1700[b]
Liquid + GdSi ↔ Gd_5Si_4	1774 (40.66)	1770[a], 1725[b]
Liquid ↔ GdSi	1835 (50.0)	1835[a], 1830[b]
Liquid + GdSi ↔ Gd_3Si_5	1664 (64.95)	1665[a], 1625[b]
Liquid + Gd_3Si_5 ↔ $GdSi_{2-x}$	1601 (71.67)	1600[a], 1590[b]
Liquid + $GdSi_{2-x}$ ↔ diamond (Si)	1226 (87.02)	1227[a], 1230 (83)[b]

[a] This work.
[b] Ref. [5].

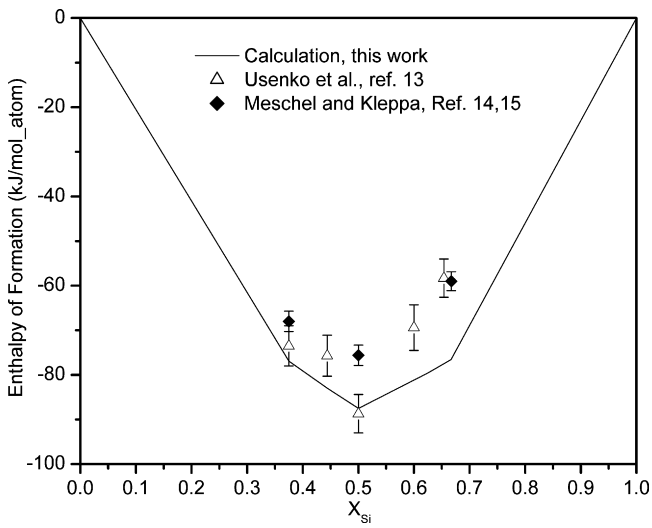

Fig. 9. The calculated enthalpy formation of the solid phases and the experimental data at 298 K. Reference state is solid Gd and Si.

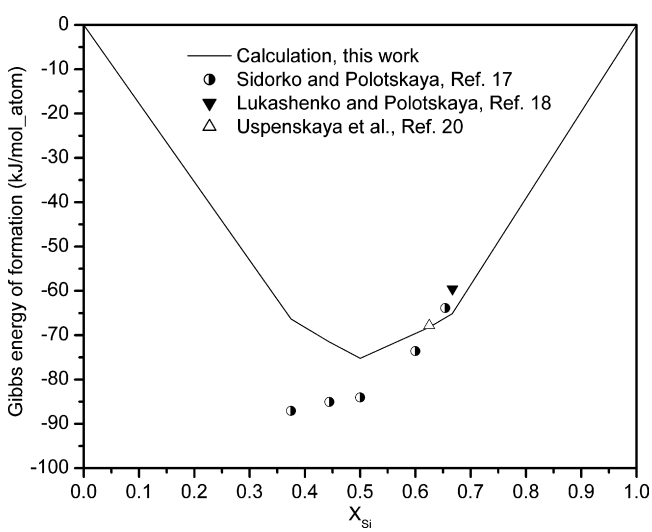

Fig. 10. The calculated Gibbs energy formation of the solid phases at 800 K and the experimental data. Reference state is solid Gd and Si.

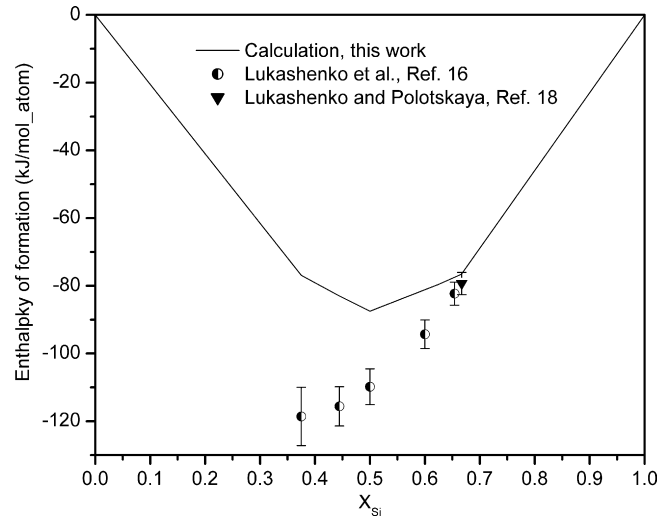

Fig. 11. The calculated enthalpy of formation of the solid phases 800 K and the experimental data. Reference state is solid Gd and Si.

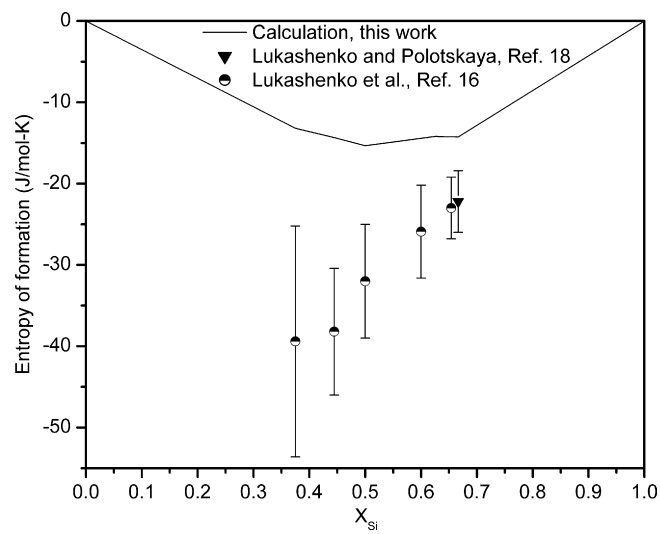

Fig. 12. The calculated entropy of formation of the solid phases at 800 K and the experimental data. Reference state is solid Gd and Si.

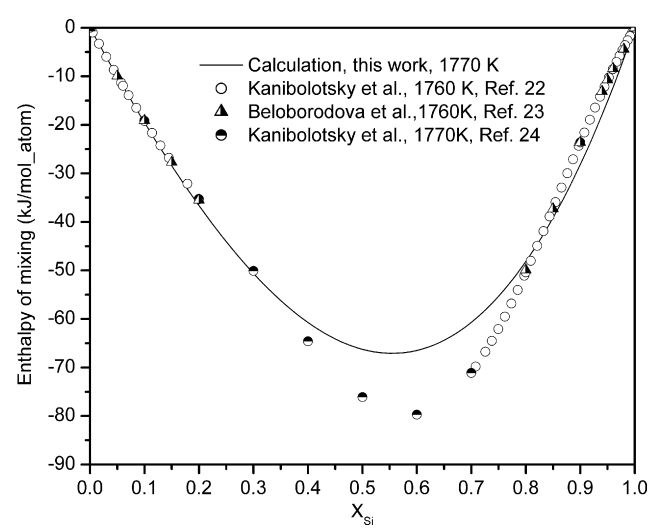

Fig. 13. The enthalpy mixing of the liquid phase at 1770 K and the experimental data.

ment with the directly measured experimental data but larger difference was observed between the calculated results and the extrapolated data in the composition range of 40–70 at.% Si.

5. Conclusions

Most of the invariant reaction temperatures are confirmed and in good agreement with reported data by thermal analysis. However, the reported infection point in the liquidus curve near 60 at.% Si was not found. Differential scanning calorimetry results indicate that the transition temperature of α-$GdSi_{2-x}$ to β-$GdSi_{2-x}$ is lower than reported data on either side of the phase boundary. XRD and SEM data indicate the composition for the compound with AlB_2 structure is Gd_3Si_5 instead of Gd_2Si_3. A thermodynamic description of the Gd–Si binary system was obtained by using the CALPHAD technique and WinPhad software. The comparison shows that the computed phase diagram and thermodynamic properties are in good agreement with the experimental data.

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